SafeDrug: A Benchmark Dataset for Safety-Critical Pharmacological Reasoning in LLMs

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Abstract

Large language models (LLMs) have shown strong performance on biomedical tasks, yet evaluating their reasoning in safety-critical pharmacological contexts requires large-scale, structured datasets. We present SafeDrug, a benchmark dataset for systematic evaluation of pharmacological reasoning across multiple task dimensions. SafeDrug integrates heterogeneous sources, including adverse drug event records, drug–drug interaction knowledge, literature-derived evidence, and population-specific information such as child, adult, and older adult groups, into a unified reasoning-oriented QA format. The benchmark comprises two complementary subsets. **SafeDrug-large** provides broad coverage at scale, while **SafeDrug-small** is a high-quality human-annotated subset for precise evaluation. It supports tasks such as prediction, risk assessment, mechanism explanation, evidence-grounded QA, and drug replacement across both single-drug and multi-drug settings. Our evaluation shows that LLMs capture outcome-level associations but struggle with mechanistic reasoning, evidence alignment, and population-aware predictions. SafeDrug provides a foundation for reproducible benchmarking and advances research on evidence-grounded and population-aware pharmacological reasoning in LLMs. The code and data are available at: https://anonymous.4open.science/r/SafeDrug-0544

1 Introduction

Large language models (LLMs) have recently demonstrated strong performance across a wide range of biomedical and clinical tasks [1, 2], including medical question answering [3, 4], literature understanding [5], and knowledge retrieval [6]. These capabilities suggest the potential of LLMs to support real-world healthcare applications. However, many of these tasks primarily evaluate factual recall or short-form reasoning, leaving open the question of whether LLMs can reliably perform *safety-critical pharmacological reasoning*, such as assessing adverse drug events (ADEs [7]), identifying drug–drug interactions (DDIs [8, 9]), and providing mechanism-grounded explanations.

Pharmacological safety assessment is fundamentally more challenging than standard biomedical QA [1], which requires multi-step reasoning over heterogeneous knowledge sources [10, 11, 12], including drug properties, biological mechanisms, and clinical evidence. In addition, real-world scenarios often involve *compositional complexity*, such as multi-drug interactions, and *population heterogeneity*, where risk profiles differ across patient groups (e.g., children, adults, and older adults).

*Equal contribution. The project page is available at: https://loop113.github.io/safedrug-benchmark

40th Conference on Neural Information Processing Systems (NeurIPS 2026) Track on Datasets and Benchmarks.

Benchmark	Domain	Reasoning	Long Chain	Mechanistic	Population	Evidence	Therapy
MedQA [20]	General Medicine	1-step QA	✗	✗	✗	✗	✗
PubMedQA [21]	Biomedical	1-step QA	✗	✗	✗	✓	✗
BioASQ [22]	Biomedical	1-step QA	✗	✗	✗	✓	✗
DrugBank-QA [10]	Discovery	1-step QA	✗	✓	✗	✗	✗
DDI-Ben [13]	Interaction	Classification	✗	✗	✗	✗	✗
SafeDrug	Drug safety	$\geq$2-step QA	✓	✓	✓	✓	✓

Table 1: Comparison of SafeDrug with existing biomedical and drug safety benchmarks. SafeDrug differs from prior benchmarks by jointly supporting multi-step pharmacological reasoning, mechanism-level evaluation, population-aware assessment, evidence alignment, and therapy-level decision tasks. ✓ indicates that the capability is supported, while ✗ indicates that it is not supported.

Errors in such settings may lead to incorrect or unsafe clinical recommendations, making reliability and interpretability essential. Despite growing interest in evaluating LLMs in the biomedical domain, existing benchmarks remain insufficient for assessing these capabilities [13, 7, 14]. Most prior datasets focus on single-step question answering or classification tasks [7, 15] and lack key aspects required for safety-critical reasoning [13], such as mechanism-level supervision [16], explicit evidence grounding, and population-aware evaluation. As a result, it remains unclear whether current LLMs genuinely understand pharmacological mechanisms or merely rely on superficial correlations in data.

To address this gap, we introduce **SafeDrug**, a comprehensive benchmark for evaluating pharmacological safety reasoning in large language models. SafeDrug organizes tasks into a hierarchical structure spanning prediction and reasoning, including ADE prediction [17], DDI detection [18, 19], risk assessment, mechanism explanation, evidence-grounded question answering, and drug replacement. The benchmark integrates heterogeneous pharmacological datasets from multiple sources and unifies them into a reasoning-oriented QA format. Importantly, SafeDrug explicitly incorporates *population-specific evaluation* and supports *multi-drug scenarios*, enabling a more realistic assessment of model behavior in complex clinical settings.

Using SafeDrug, we conduct a systematic evaluation of state-of-the-art LLMs and uncover several key limitations. First, while models achieve competitive performance on prediction tasks, they struggle with mechanism explanation and evidence-grounded reasoning, revealing a gap between knowledge memorization and mechanistic understanding. Second, performance degrades significantly with increasing reasoning complexity, particularly in multi-drug settings, indicating limited compositional reasoning ability. Third, models often fail to faithfully utilize supporting evidence, producing hallucinated or weakly grounded justifications. Finally, we observe substantial performance disparities across patient populations, suggesting that current models lack robustness in underrepresented groups.

Our work highlights fundamental challenges in applying LLMs to safety-critical pharmacological tasks and provides a benchmark for systematically evaluating and understanding these limitations. We summarize our main contributions as follows: (1) We introduce **SafeDrug**, a comprehensive benchmark dataset for evaluating safety-critical pharmacological reasoning in large language models, consisting of a large-scale collection (SafeDrug-large) and a high-quality human-annotated subset (SafeDrug-small). (2) We design a hierarchical task framework spanning prediction, risk assessment, mechanistic reasoning, evidence grounding, and decision-making under multi-drug and population-specific settings. (3) We conduct a large-scale evaluation of state-of-the-art LLMs, revealing systematic gaps between outcome-level prediction and deeper reasoning, particularly in mechanism understanding, evidence alignment, and population robustness. (4) We provide detailed diagnostic analyses, including capability breakdown and error taxonomy, to support future research on reliable, interpretable, and population-aware pharmacological reasoning.

2 Benchmarking Adverse Drug Events with SafeDrug

We introduce **SafeDrug**, a comprehensive benchmark for evaluating safety-critical pharmacological reasoning in LLMs. As illustrated in Figure 1, SafeDrug is designed to move beyond conventional prediction-based evaluation by incorporating structured reasoning, evidence grounding, and population-aware decision-making into a unified framework.

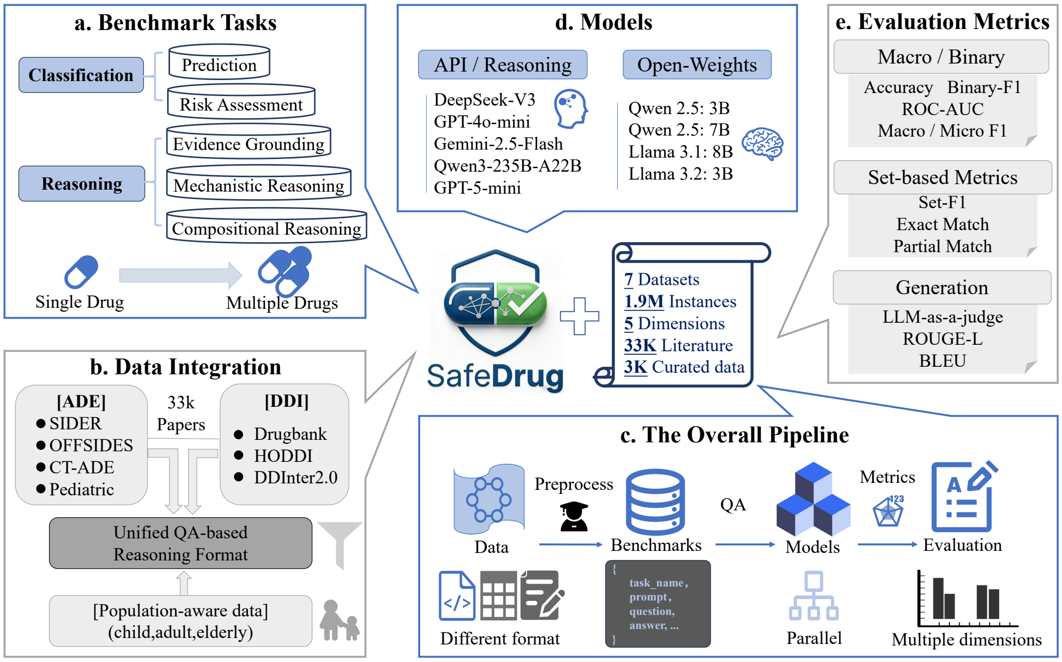

Figure 1: SafeDrug is a benchmark for evaluating safety-critical pharmacological reasoning in large language models. It covers a hierarchy of tasks from prediction to multi-step reasoning (a), integrates heterogeneous ADE/DDI data with population awareness into a unified QA format (d), and evaluates diverse LLMs using multiple metrics for both accuracy and reasoning quality (b–c). The pipeline standardizes data, performs model inference, and conducts multi-dimensional evaluation (e).

2.1 Benchmark Tasks

SafeDrug organizes pharmacological tasks into a hierarchical structure that spans *classification* and *reasoning* (Figure 1a). This design reflects the increasing complexity of drug safety assessment, from outcome prediction to structured, multi-step reasoning.

Classification Tasks. At the base level, SafeDrug includes outcome-oriented prediction tasks: *(i) prediction* and *(ii) risk assessment*. The prediction task focuses on identifying potential adverse drug events (ADEs) or interactions, while risk assessment requires estimating the severity or likelihood of such events. These tasks primarily evaluate whether models can capture statistical associations between drugs and clinical outcomes. However, they do not require an understanding of underlying mechanisms or justification of predictions.

Reasoning Tasks. To evaluate deeper pharmacological understanding, SafeDrug introduces reasoning-centric dimensions such as *evidence grounding*, which requires models to support predictions with relevant evidence, ensuring both correctness and justification. This is critical in safety-sensitive settings, where unsupported outputs may lead to unreliable conclusions. To assess this capability, we design two complementary tasks: evidence selection and evidence generation. *(iv) mechanistic reasoning* evaluates whether models can explain drug effects and interactions based on biological or pharmacological mechanisms (e.g., metabolism or toxicity pathways). This dimension goes beyond correlation to assess mechanistic understanding. To this end, we include both mechanistic classification and generation tasks. *(v) compositional reasoning* focuses on multi-drug scenarios, where interactions emerge from combinations of drugs. These tasks require models to reason over multiple entities and their interactions for prediction and risk assessment, reflecting real-world settings such as polypharmacy. Together, these dimensions define a progression from surface-level prediction to structured reasoning. In particular, SafeDrug explicitly models the transition from *single-drug* to *multi-drug* settings, increasing reasoning complexity and enabling systematic evaluation of model capabilities under realistic pharmacological conditions.

2.2 Data Integration

SafeDrug integrates heterogeneous pharmacological data sources into a unified reasoning-oriented format (Figure 1b). More details about preprocessing and QA templates in Appendix A.3 and A.4.

ADE Data for Single Drug. We incorporate large-scale adverse drug event (ADE) datasets, including SIDER [23], OFFSIDES [24], CT-ADE [7], and pediatric-specific resources [25]. These datasets provide drug–adverse effect associations, frequency statistics, and clinical reports, capturing relationships between drug exposure and observed outcomes. ADE data primarily support *prediction* and *risk assessment* tasks, offering a statistical foundation for identifying potential adverse events. However, such associations do not encode causal mechanisms or supporting evidence, and thus mainly assess the knowledge recall capability of LLMs rather than their reasoning ability.

ADE Data for multiple Drugs. We incorporate curated drug–drug interaction (DDI) datasets, including DrugBank [10], HODDI [26], and DDInter2.0 [15]. These resources provide structured information on interaction types, severity levels, and, in some cases, underlying pharmacological mechanisms (e.g., metabolic pathways or transporter effects). In particular, HODDI enables evaluation of *compositional reasoning* in multi-drug settings and offers partial support for *mechanistic reasoning*. Importantly, these datasets reflect the combinatorial nature of real-world drug usage, where interactions arise from multiple agents rather than isolated drugs, and provide sufficient coverage for evaluating LLMs on both prediction and compositional reasoning tasks.

Population-Aware Data. To capture clinical heterogeneity, SafeDrug incorporates population-specific information, including pediatric, adult, and older adult groups, from pediatric-focused resources [25] and CT-ADE [7]. These data reflect age-related differences in drug response, such as variations in metabolism, organ function, and comorbidity profiles. Population-aware data enable evaluation of whether models can adapt predictions and reasoning across patient groups, supporting more realistic and safety-critical assessment scenarios.

Evidence and Literature Data. In addition to structured datasets, SafeDrug integrates large-scale literature-derived evidence from DDInter2.0, comprising approximately 33K curated biomedical documents. We further process these sources using DeepSeek-V3.2 [27] to extract and summarize adverse event mechanisms and supporting evidence, which are then organized into reasoning-oriented QA instances. To ensure data quality, we perform stratified sampling based on the number of supporting documents per event and conduct manual verification on a subset of instances (refer to Appendix A.4). This pipeline provides reliable and structured evidence for evaluating *evidence grounding* and *mechanistic reasoning*, and supports future research on LLM fine-tuning for pharmacological reasoning.

Unified QA-based Format. All data sources are unified into a *reasoning-oriented QA format*, where each instance includes a question, structured context, and target outputs (e.g., answer, mechanism, evidence). This design enables consistent evaluation across heterogeneous tasks and supports assessment of both prediction accuracy and reasoning quality. By integrating complementary data modalities, SafeDrug facilitates comprehensive evaluation of pharmacological reasoning under realistic clinical conditions. Details of the construction process and task-specific templates are provided in Appendix A.

2.3 Benchmark Construction

The construction pipeline of SafeDrug is illustrated in Figure 1c, where heterogeneous pharmacological data are transformed into a unified and comparable evaluation framework for LLMs. The overall data distribution is shown in Figure 2, with detailed statistics provided in Appendix A.

Data Standardization and QA Construction. Raw data from heterogeneous sources (e.g., structured databases and textual evidence) are first standardized via preprocessing, including entity normalization and terminology alignment (e.g., mapping drugs and adverse events to unified vocabularies). This ensures consistency across datasets and avoids spurious discrepancies caused by format or naming differences. We then convert standardized data into a *reasoning-oriented QA format* using task-specific prompt generation, where each instance is framed as a question requiring prediction, reasoning, or decision-making with structured context. This design enables a unified evaluation interface across tasks and aligns evaluation with the interactive reasoning capabilities of modern LLMs. Each instance follows a consistent schema (e.g., `task_name`, `prompt`, `question`, `answer`), supporting parallel evaluation and fine-grained analysis of prediction accuracy, reasoning quality, and evidence alignment, while allowing extension to additional outputs such as mechanisms and supporting evidence. Detailed descriptions of mechanism labels and evidence supports are provided in Appendix A.1. Details of drug replacement (or substitution) labels are provided in Appendix A.2.

Scale and Coverage. SafeDrug consists of two complementary subsets. **SafeDrug-large** is a large-scale collection built from **7 datasets**, containing approximately **1.9M instances** across **5 task dimensions**. **SafeDrug-small** is a high-quality human-curated subset obtained via stratified sampling of **3,000** instances based on supporting evidence counts, followed by expert verification, resulting in **2,554** high-confidence instances. In addition, SafeDrug integrates over **33K literature-derived evidence entries** to support evaluation of evidence grounding and mechanistic reasoning, enabling both large-scale coverage and reliable assessment of pharmacological reasoning. We provide a detailed annotation protocol for human labeling in Appendix A.4.

2.4 Evaluation Protocol

SafeDrug adopts a multi-dimensional evaluation protocol to assess both answer correctness and reasoning reliability (Figure 1e). This is necessary due to the heterogeneous nature of drug safety tasks, which range from binary decisions to set prediction and free-form mechanistic explanations. Accordingly, a single metric is insufficient to capture the full spectrum of ADE and DDI reasoning ability.

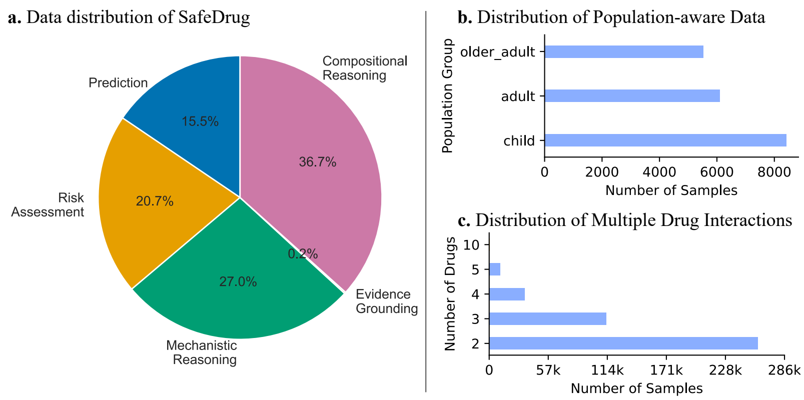

Figure 2: Data statistics of SafeDrug. (a) Task distribution across prediction, risk assessment, and reasoning dimensions. (b) Population-aware data distribution over child, adult, and older adult groups. (c) Distribution of multi-drug interactions with varying numbers of drugs, reflecting polypharmacy complexity.

Evaluation Metrics. We adopt three complementary metric families to evaluate different task types. For prediction-oriented tasks (e.g., ADE prediction, DDI detection, and risk assessment), we use classification metrics such as Accuracy, Binary-F1, Macro/Micro-F1, and ROC-AUC, where Macro-F1 is particularly important due to class imbalance in adverse events. For tasks with multiple valid outputs (e.g., ADE sets, evidence selection, and multi-drug interactions), we use set-based metrics including Set-F1, Exact Match, and Partial Match to capture both correctness and coverage. For open-ended reasoning tasks (e.g., mechanism explanation and evidence-grounded QA), we use generation metrics such as ROUGE, BLEU, and LLM-as-a-judge to assess semantic consistency, coherence, and evidence alignment. Detailed metric formulations are provided in Appendix B.1.

2.5 Model Coverage

SafeDrug evaluates a diverse set of large language models, including both API-based (closed-source LLMs) and open-weight models (Figure 1d). This design is intended to assess whether safety-critical pharmacological reasoning is a general capability of current LLMs or primarily emerges in the strongest proprietary systems.

API-based Models. We include state-of-the-art API-based models, which typically represent the strongest available general-purpose and reasoning-oriented LLMs. Evaluating these models provides an estimate of the current upper bound of LLM performance on drug safety reasoning tasks. In

Table 2: Performance of representative LLMs on SafeDrug single-drug and pairwise DDI tasks. Exact Match scores show that models perform better on prediction tasks than on reasoning tasks, highlighting limitations in mechanistic understanding.

Model	**ADE**		**Drug Interaction**		**Reasoning**	
	Prediction	Risk	Prediction	Risk	Drug Replacement	Mechanism Prediction
Qwen2.5:7B	15.73	29.53	28.16	42.55	68.42	23.28
Qwen2.5:3B	13.90	12.39	21.42	29.31	60.34	42.49
LLaMA3.1:8B	3.06	32.94	28.96	47.15	66.40	4.22
LLaMA.2:3B	14.78	12.85	29.82	40.36	59.40	5.02
DeepSeek-V3	23.15	29.77	26.96	35.12	59.90	29.23
GPT-4o-mini	13.49	25.49	29.29	47.43	68.06	31.04
Gemini-2.5-flash	10.90	38.10	63.55	35.60	78.85	21.70
Qwen3-235B	11.05	25.10	32.55	29.00	75.70	27.00
GPT-5-mini	8.25	27.00	57.60	48.90	78.80	24.60

Table 3: Performance of representative LLMs on SafeDrug polypharmacy tasks.

Model	**Multiple Drug Prediction**				**Multiple Drug Risk**			
	3 Drugs	4 Drugs	5 Drugs	10 Drugs	3 Drugs	4 Drugs	5 Drugs	10 Drugs
Qwen2.5:7B	47.11	48.60	53.13	65.29	42.83	38.66	39.27	45.24
Qwen2.5:3B	57.60	63.30	66.42	69.42	68.33	74.71	84.83	95.24
LLaMA3.1:8B	41.25	45.05	52.30	43.80	37.39	32.25	28.20	21.43
LLaMA3.2:3b	40.32	51.94	61.64	65.29	58.93	58.42	62.45	66.67
DeepSeek-V3	60.34	64.59	65.76	66.11	63.98	69.96	76.04	91.66
GPT-4o-mini	43.74	48.83	52.08	51.23	46.21	45.57	46.42	57.14
Gemini-2.5-flash	62.60	65.05	65.90	66.11	65.20	71.60	80.20	94.04
Qwen3-235B	56.75	59.80	62.60	64.46	61.20	67.45	70.45	89.28
GPT-5-mini	58.65	63.45	65.05	66.11	50.35	53.05	56.65	69.04

particular, they allow us to examine whether high-capacity models can reliably perform mechanism explanation, evidence-grounded reasoning, and population-specific ADE assessment.

Open-weight Models. We also evaluate open-weight models to ensure reproducibility and to understand how accessible models perform under the same benchmark protocol. These models are important for analyzing whether pharmacological reasoning failures are shared across model families, and whether smaller or openly available systems exhibit larger gaps in evidence use, mechanism reasoning, and multi-drug compositionality.

3 Experiment

Additional implementation details and experimental results on SafeDrug are provided in Appendix B.

3.1 Experimental Setup

Datasets. We aggregate samples from multiple sources to construct a unified SafeDrug collection with 1.9M instances, split into train, validation, and test sets at an 8:1:1 ratio. Additionally, we create a high-precision subset of 2,554 instances (train/val/test = 1,817/331/406) for focused benchmarking. All experiments are conducted on held-out test splits to evaluate LLM performance across prediction, risk assessment, reasoning, and evidence-grounded tasks.

Models. We benchmark a representative set of nine large language models, including both open-weight and proprietary systems. Open-weight models include `Qwen2.5-7B` [28], `Qwen2.5-3B` [28], `LLaMA-3.1-8B` [29], and `LLaMA-3.2-3B` [29]. Proprietary and frontier models include `DeepSeek-V3` [30], `GPT-4o-mini` [31], `Gemini-2.5-Flash` [32], `Qwen3-235B-A22B` [33], and `GPT-5-mini` [34].

Implementation Details. All experiments were conducted in a **zero-shot** setting with deterministic decoding ($T = 0$) to isolate model reasoning capabilities without task-specific fine-tuning. Open-source models were evaluated on a server with a single NVIDIA RTX 3080Ti using the

Ollama framework, while closed-source models were accessed via proxy APIs (ChatAnyWhere[2]). For multiple-choice tasks, answer options were shuffled using a fixed random seed (42) to mitigate positional biases. Model outputs were post-processed using pattern matching to extract final predictions and structured fields such as predicted labels, supporting evidence, and mechanistic annotations when applicable. This setup enables consistent comparison across models and tasks while controlling for variability introduced by decoding stochasticity or formatting differences. Full details of preprocessing, prompt design, and post-processing are provided in Appendix A.

3.2 Overall Performance on SafeDrug

We first evaluate representative LLMs on single-drug and pairwise DDI tasks using Exact Match (EM) scores, as summarized in Table 2. The evaluation covers three dimensions:*ADE prediction and risk assessment*, *DDI prediction and risk assessment*, and reasoning tasks including *drug replacement* and *mechanism prediction*. From the reported results, we have there main observations: (1) Across models, we observe that LLMs generally achieve higher performance on outcome-level prediction tasks, such as ADE and DDI prediction, compared with reasoning-oriented tasks. For instance, models like GPT-5-mini and Gemini-2.5-Flash reach EM scores above 60% on DDI prediction, while mechanism prediction remains below 30% for most models. This indicates that current LLMs are capable of capturing statistical associations between drugs and adverse events, but struggle to provide mechanistic explanations that require deeper pharmacological understanding. (2) Similarly, risk assessment tasks present a substantial challenge. Many models, including Qwen2.5 variants and LLaMA3 models, exhibit a marked drop in EM scores when predicting severity or risk labels compared with straightforward ADE prediction. This suggests that LLMs may rely on surface-level correlations rather than accurately reasoning about clinical severity. (3) Reasoning tasks such as drug replacement and mechanism prediction further reveal the limitations of current models. Even high-capacity models like GPT-5-mini achieve modest performance on mechanism prediction (24.6%) despite strong drug replacement scores (78.8%), highlighting a gap between prediction capability and mechanistic reasoning. These results underscore that SafeDrug effectively probes both predictive and reasoning capabilities, revealing systematic weaknesses in mechanistic and evidence-grounded understanding in current LLMs. Overall, the findings demonstrate that while LLMs can perform reasonably on single-step prediction tasks, they remain limited on tasks requiring multi-step reasoning, evidence alignment, and mechanistic understanding, motivating further research on improving safety-critical pharmacological reasoning.

3.3 Capability Breakdown: Prediction vs. Reasoning

To gain a more granular understanding of LLM strengths and limitations on SafeDrug, we evaluate model performance across five key capability dimensions: prediction, risk assessment, mechanistic reasoning, evidence grounding, and compositional reasoning. For each model, we compute task-specific metrics (e.g., Exact Match, Set-F1, LLM-as-a-judge scores) and aggregate them by capability to visualize relative performance, as shown in the radar plot in Figure 3. The analysis reveals systematic differences between outcome-level prediction and reasoning-oriented capabilities. Most models achieve their highest scores on prediction and compositional reasoning, indicating that LLMs can capture statistical patterns in drug-ADE associations and multi-drug correlations. By contrast, mechanistic reasoning and evidence grounding consistently exhibit lower performance, even for high-capacity models such as GPT-5-mini and Gemini-2.5-Flash. This discrepancy highlights a gap between superficial pattern recognition and

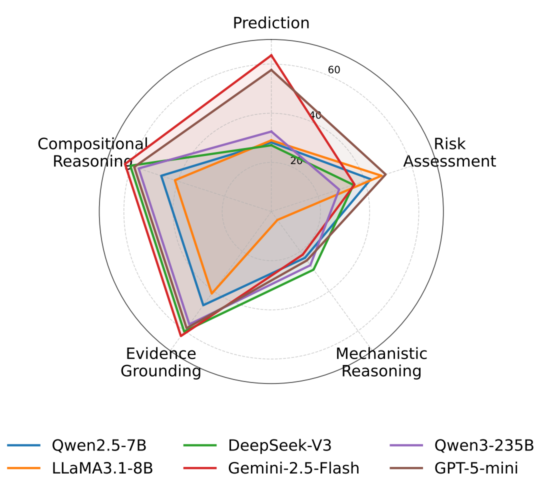

Figure 3: Capability breakdown of representative LLMs on SafeDrug.

[2]https://api.chatanywhere.tech

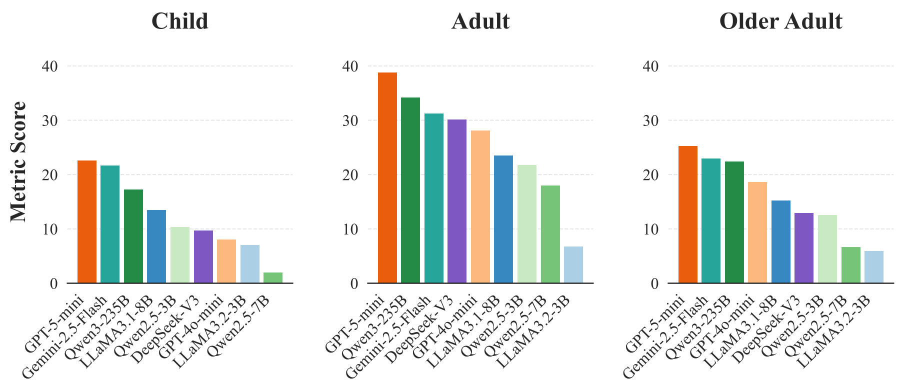

Figure 4: Performance comparison (Set-F1) of evaluated LLMs on population-specific ADE prediction (Child, Adult, Older Adult).

deeper mechanistic understanding. We also observe that risk assessment scores are intermediate: models can often identify interactions, but fail to accurately assess their severity. The radar plot further shows that compositional reasoning benefits from increased context, yet performance on mechanistic and evidence-alignment tasks remains bottlenecked. Collectively, these results suggest that current LLMs possess robust outcome-prediction capabilities but are limited in faithful mechanistic and evidence-based reasoning. Overall, this capability breakdown underscores the necessity of multi-dimensional benchmarks like SafeDrug, which not only evaluate predictive accuracy but also probe reasoning fidelity, evidence use, and compositional generalization under complex pharmacological scenarios.

3.4 Compositional Reasoning under Polypharmacy

We evaluate LLM performance on multi-drug prediction and risk assessment tasks to probe compositional reasoning under polypharmacy, where interactions among multiple drugs can give rise to complex adverse events. Exact Match scores are reported for 3, 4, 5, and 10 drug combinations, as summarized in Table 3 and visualized in Figure 7 (Appendix B.1.3). Across models, we observe that prediction accuracy generally increases or remains stable as the number of drugs grows, indicating that models can capture surface-level associations between drugs and potential interactions. For example, Gemini-2.5-Flash and GPT-5-mini achieve over 65% Exact Match on 10-drug prediction, demonstrating strong outcome-level pattern recognition. However, risk assessment performance is more variable and often lower than prediction, highlighting that models struggle to correctly estimate interaction severity as combinatorial complexity increases. Some models over-predict high-risk labels when the number of drugs is large, suggesting reliance on heuristics rather than explicit reasoning over specific interactions. These results indicate that while current LLMs can handle outcome-level multi-drug predictions, they remain limited in compositional reasoning, particularly in assessing risk. The findings underscore the need for benchmarks like SafeDrug that explicitly probe multi-drug interactions and compositional reasoning, beyond simple predictive associations.

3.5 Population-Specific ADE Evaluation

To assess model robustness across different patient populations, we evaluate LLM performance on ADE prediction for children (< 18 years old), adults (18–64 years old), and older adults (≥ 65 years old), as shown in Figure 4. This analysis is critical because pharmacological responses can vary

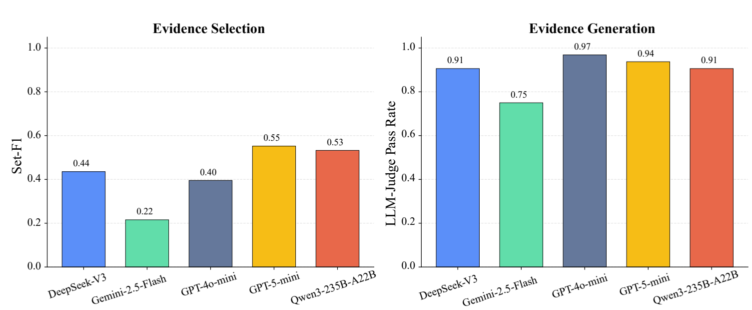

Figure 5: Evaluation of LLM performance on evidence tasks in SafeDrug.

with age, and overrepresentation of adult data in existing sources may bias model predictions. The results reveal a consistent adult-centric bias across most models. Models generally achieve higher metric scores on adult instances, with GPT-5-mini and Gemini-2.5-Flash reaching the highest performance. In contrast, performance on pediatric and older adult populations is substantially lower, particularly for smaller open-source models such as Qwen2.5-7B and LLaMA3.2-3B. This indicates that models tend to rely on patterns learned from adult-centric data and fail to generalize effectively to underrepresented populations. These findings highlight the importance of including population-stratified evaluation in benchmarks for safety-critical pharmacological reasoning. Without such assessment, models may appear performant overall while exhibiting systematic biases that could compromise reliability in clinical settings for children and older adults.

3.6 Evidence Grounding and Faithfulness

We evaluate LLM performance on evidence-grounded tasks in SafeDrug, focusing on two dimensions: *evidence selection*, measured by Set-F1, and *evidence generation*, assessed via LLM-as-a-judge pass rate (Figure 5). Results indicate that models achieve moderate performance on evidence selection, with Set-F1 scores ranging from 0.22 (Gemini-2.5-Flash) to 0.55 (GPT-5-mini), suggesting that models often fail to reliably retrieve all relevant supporting information. Evidence generation performance is higher, with LLM-as-a-judge pass rates exceeding 0.9 for most models, indicating that when generating textual explanations, models can produce mostly plausible and coherent evidence. These findings reveal a gap between retrieval and generative reasoning: LLMs can produce convincing explanations, but they frequently omit or misalign supporting evidence. This highlights the need for benchmarks like SafeDrug that probe not only predictive accuracy but also the faithfulness of mechanistic and evidence-grounded reasoning.

3.7 Error Taxonomy Analysis

We analyze LLM failures across SafeDrug task categories using a taxonomy of five error types: context misalignment, missed signals (false negatives), hallucination/fabrication, over-risk/exaggeration, and superficial reasoning (Figure 6). The analysis reveals distinct failure patterns depending on the task. Prediction and risk assessment errors are dominated by missed signals and over-risk estimates, indicating that models often fail to capture rare events or overestimate severity. In contrast, evidence grounding and mechanistic reasoning tasks exhibit higher rates of context misalignment and hallucinated mechanisms, reflecting difficulties in aligning model outputs with supporting evidence and biological mechanisms. Compositional reasoning errors combine hallucination and missed signal effects, highlighting the challenges of multi-drug interactions. These findings demonstrate that SafeDrug effectively exposes systematic limitations in LLM reasoning, distinguishing superficial pattern recognition from mechanistic and evidence-grounded reasoning failures.

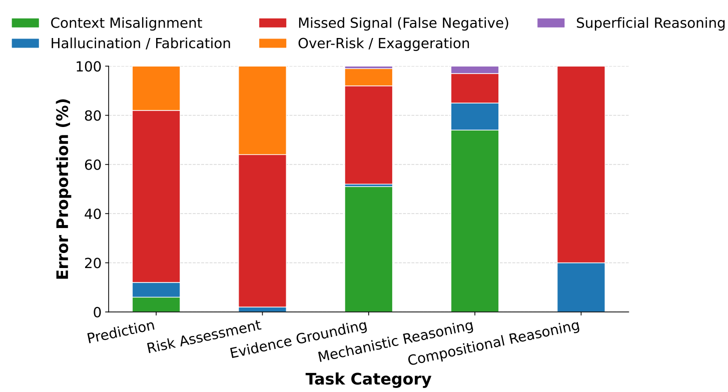

Figure 6: Errors are categorized into context misalignment, missed signals, hallucination/fabrication, over-risk/exaggeration, and superficial reasoning. Prediction and risk assessment errors are dominated by missed signals and over-risk estimates, whereas mechanistic and evidence-grounded reasoning tasks show higher rates of context misalignment and hallucinated mechanisms.

4 Discussion

Our experiments on SafeDrug show that LLMs perform well on outcome-level prediction but struggle with mechanistic reasoning, evidence grounding, risk assessment, and population-sensitive tasks, revealing a gap between surface associations and evidence-based reasoning for clinical safety. Future work should improve mechanistic and compositional reasoning, enhance evidence use, and address population biases via better data and modeling strategies. Current limitations include underrepresented subpopulations and incomplete evidence annotations, motivating expanded coverage and richer mechanistic data. Overall, SafeDrug provides a structured benchmark to advance robust and interpretable pharmacological reasoning in LLMs.

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Technical Appendix

A Details of Data Preprocessing

A.1 Mechanism and Evidence Supervision

Mechanism labels and evidence rationales were not collected through large-scale manual annotation from scratch. Instead, we first derived supervision from DDInter 2.0 itself, including interaction descriptions, management recommendations, reference lists, and alternative-drug fields. After evidence-aware sampling, all selected source interactions were manually checked by our team for pair consistency, task assignment, label validity, and evidence quality. Thus, the benchmark is best characterized as *source-grounded with manual verification*, rather than fully human-authored. At the benchmark level, all non-generative supervision is source-derived rather than LLM-generated. This includes classification, multi-label, multi-choice, and span-selection targets. In our cleaned release, this accounts for 2,154 of 2,554 instances ($\approx 84.3\%$). LLM assistance is used only for generative reference fields, namely mechanism reasoning and evidence reasoning, which together account for 400 of 2,554 instances ($\approx 15.7\%$). These fields are generated from DDInter text and, when available, literature-enriched abstracts, and are treated as reference answers rather than hidden labels for structured prediction. To guard against circularity, we adopt three safeguards. First, LLM-synthesized fields are never used as gold labels for objective or semi-structured tasks. Second, the released `model_input` hides answer-bearing fields for objective tasks, reducing shortcut access to supervision. Third, generative evaluation is reported separately, and LLM-based judging is performed with a fixed external judge rather than using the structured labels themselves. As a result, LLM assistance serves only as a controlled augmentation for free-text reference construction, not as a source of circular supervision for the main structured benchmark.

A.2 Therapy-level Drug Replacement and Clinical Plausibility

For therapy-level drug replacement (or substitution) tasks, SafeDrug follows a structured and clinically motivated construction strategy based on DDInter2.0. Specifically, replacement candidates are derived using the Anatomical Therapeutic Chemical (ATC) classification system [35], where drugs sharing the same **fourth-level ATC code** are considered therapeutically equivalent. This level of the ATC hierarchy corresponds to pharmacological or therapeutic subgroups, ensuring that candidate replacements have similar mechanisms of action and clinical indications.

This design enables scalable and consistent annotation of replacement options while maintaining a reasonable degree of clinical plausibility. Compared to heuristic or model-inferred substitutions, ATC-based grouping provides a standardized and widely accepted framework for identifying alternative drugs within the same therapeutic class. For SafeDrug-small, all replacement candidates are further manually verified by domain experts. The verification process ensures that suggested alternatives are clinically plausible, do not introduce obvious safety risks, and are consistent with the context of the interaction. Ambiguous or inappropriate substitutions are removed during this process.

We emphasize that these replacement annotations are intended for *benchmarking and evaluation purposes only*. They approximate therapy-level alternatives based on pharmacological similarity, but do not constitute clinical treatment recommendations. Real-world drug substitution requires comprehensive consideration of patient-specific factors, comorbidities, and clinical guidelines.

A.3 Details of SafeDrug-large

The **SafeDrug-large** benchmark is systematically designed to evaluate Large Language Models (LLMs) on diverse and complex pharmacovigilance tasks. In this section, we detail the demographic categorization principles and provide concrete task formulations through QA examples.

A.3.1 Population Demographics and Data Sources

A key differentiator of SafeDrug-large is its fine-grained demographic stratification, explicitly integrating patient age into Adverse Drug Event (ADE) predictions.

To construct the age-specific subset, we primarily utilize data from the **CT-ADE** and **Pediatric** databases. We standardize the population into three distinct age categories: *child*, *adult*, and *older_adult*. The categorization follows these strict rules:

- **CT-ADE Mapping:** The inherent age attributes provided in the CT-ADE database naturally align with our three categories (*child, adult, older_adult*), which we directly adopt.
- **Pediatric Database Mapping:** All records sourced from the specialized Pediatric dataset are systematically mapped and merged into the *child* category.

To preserve this rich metadata, our global integrated database (`Single_Drug_Global_Data.jsonl`) organizes demographic attributes under a comprehensive `demographics` key. This structured schema contains several sub-attributes, including `age_group`, `gender`, `is_healthy_volunteer`, and `is_pediatric_specific`. This granular structural design ensures that downstream tasks accurately reflect demographic-aware pharmacological reasoning.

A.3.2 Task Formulation and QA Examples

To clearly illustrate the evaluation formats, we provide representative JSON-formatted QA examples for each fundamental task in the SafeDrug-large benchmark. All prompts explicitly instruct the LLMs to strictly follow output constraints to facilitate automated evaluation.

1. Age-Specific ADE Prediction
Evaluates ADE identification constrained by a specific age demographic (e.g., Adult).

Example from ade_prediction_age/adult.jsonl

```
{
  "task_name": "Adult_SideEffect_DBQA",
  "prompt": "You are an expert clinical pharmacologist... \n[SCENARIO:
     Adult Case Study]\nPatient Profile: Adult Patient (18-64 years)
    .\nDrug Prescribed: Ulevostinag\nTask: Identify side effects
    that are specifically documented or observed in adult
    populations for this drug.\nOptions:\nA. Infusion site
    thrombosis\nB. Injection site reaction\n...\nF. None of the
    above\nInstructions: Provide ONLY the option letters.",
  "answer": "B, E"
}
```

2. ADE Risk and Frequency Prediction
Predicts the occurrence frequency (CIOMS definitions) of an ADE based on SIDER and OFFSIDES.

Example from ade_risk_prediction/sider/sider_risk_dataset.jsonl

```
{
  "task_name": "SIDER_Frequency_Classification",
  "prompt": "Drug Prescribed: montelukast... \nAdverse Drug Event (ADE
    ): Hypersensitivity\nTask: Predict the occurrence frequency
    category of this specific ADE.\nOptions:\nA. Very common (>=
    10%)\nB. Common (1% to < 10%)\nC. Uncommon (0.1% to < 1%)\nD.
    Rare (0.01% to < 0.1%)\nE. Very rare (< 0.01%)",
  "answer": "D"
}
```

3. Dual-Drug Interaction (DDI) Type Prediction
Identifies the primary pharmacological mechanism or type of interaction between two drugs.

Example from ddi_prediction/ddi_prediction.jsonl

```
{
  "task_name": "DDI_Type_Prediction",
  "prompt": "[SCENARIO: Drug-Drug Interaction Assessment]\nDrug A:
    Methylergometrine\nDrug B: Ciprofloxacin\nTask: Identify the
    primary pharmacological interaction type or mechanism.\nOptions
```

```
      :\nA. Others\nB. Synergy\nC. Unknown\nD. Excretion\nE.
      Metabolism",
  "answer": "E"
}
```

4. Dual-Drug Interaction Risk Assessment
Predicts the clinical severity level of a specific dual-drug interaction.

Example from ddi_risk_prediction/ddi_risk_prediction.jsonl

```
{
  "task_name": "DDI_Severity_Prediction",
  "prompt": "Drug A: Succinylcholine\nDrug B: Nitrous oxide\
      nInteraction Type: Synergy\nTask: Predict the clinical severity
      (risk level) of this specific type of interaction.\nOptions:\nA.
       Major\nB. Minor\nC. Moderate",
  "answer": "C"
}
```

5. Polypharmacy (Multi-Drug) Interaction
Determines whether concurrent administration of multiple drugs (e.g., 3, 4, 5, or 10) causes significant ADEs.

Example from multiple_drug_prediction/ddi_prediction/3_drug.jsonl

```
{
  "task_name": "Polypharmacy_Interaction_Prediction_3_Drugs",
  "prompt": "Drugs Prescribed Concurrenty:\n1. Dasabuvir\n2. Isopropyl
       alcohol\n3. Ribociclib\nTask: Predict whether the concurrent
      administration of all these specific drugs together leads to a
      known significant adverse drug-drug interaction (DDI).\nOptions
      :\nYes\nNo",
  "answer": "No"
}
```

A.4 Details of SafeDrug-small

Our benchmark is constructed from DDInter 2.0 through a multi-stage pipeline. We start with 277K interaction records containing structured fields such as descriptions, management recommendations, references, and alternative drugs, and select 200 evidence-rich interactions via stratified sampling based on an evidence-oriented score incorporating reference density, severity, management completeness, and mechanistic cues. All sampled instances are manually verified to ensure consistency of interaction pairs, labels, and evidence, and ambiguous or noisy samples are removed. From each verified interaction, we construct a structured evaluation set spanning three task tiers: objective (e.g., interaction check, severity classification), semi-structured (e.g., evidence and mechanism span selection), and generative (e.g., mechanism and evidence reasoning), with task-dependent context to avoid label leakage.

To further ensure reliability, we construct SafeDrug-small via stratified sampling of 3,000 instances from SafeDrug-large based on evidence density, followed by expert verification by a team consisting of a clinical practitioner (Master-level), a Ph.D. researcher in AI for health, and a professor specializing in AI for biological systems. The verification follows concise guidelines:

```
[Guidelines]
(1) Interaction correctness: check drug pair, type, and severity consistency.
(2) Evidence alignment: ensure references support the interaction.
(3) Mechanism validity: confirm biological plausibility.
(4) Quality control: remove ambiguous, inconsistent, or low-confidence cases.
```

This process yields 2,554 high-confidence instances. We further augment `interaction_check` with synthetic hard negatives to enable true binary evaluation. The final benchmark contains 2,554

Table 4: Sample counts for each task in SafeDrug-large.

Task	Samples
ADE Prediction (Age)	20,040
ADE Prediction (Overall)	38,934
ADE Risk Prediction (OFFSIDES)	67,452
ADE Risk Prediction (SIDER)	70,290
DDI Prediction	260,060
DDI Risk Prediction	260,060
DDI Reasoning: Counterfactual	1,076
DDI Reasoning: Mechanism Generation	260,060
DDI Reasoning: Mechanism Prediction	260,060
DDI Reasoning: Substitution Prediction	415,413
Multiple-Drug DDI Prediction (3 drugs)	84,376
Multiple-Drug DDI Prediction (4 drugs)	35,116
Multiple-Drug DDI Prediction (5 drugs)	11,508
Multiple-Drug DDI Prediction (10 drugs)	601
Multiple-Drug Risk Prediction (3 drugs)	113,595
Multiple-Drug Risk Prediction (4 drugs)	34,413
Multiple-Drug Risk Prediction (5 drugs)	10,474
Multiple-Drug Risk Prediction (10 drugs)	417
Total	**1,943,945**

instances, split by source interaction to prevent data leakage (1,817/331/406 for train/dev/test). The released `split-clean` version retains only evaluation-critical fields, including `model_input`, `label`, `answer`, and `scoring_hint`, making it directly usable for controlled evaluation.

B Additional Experiments

B.1 Evaluation Protocol

In this section, we provide the formal mathematical definitions for the three metric families utilized in our evaluation framework.

B.1.1 Classification Metrics (Prediction-oriented Tasks)

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ be the dataset with N samples, where $y_i \in \mathcal{C}$ is the ground truth label and $\hat{y}_i$ is the predicted label. For a specific class c, we denote True Positives (TP_c), False Positives (FP_c), True Negatives (TN_c), and False Negatives (FN_c).

Accuracy and F1-Scores. Accuracy measures the overall proportion of correct predictions:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(\hat{y}_i = y_i) \tag{1}$$

where $\mathbb{I}(\cdot)$ is the indicator function. For a specific class c (or binary classification), Precision (P_c) and Recall (R_c) are defined as $P_c = \frac{TP_c}{TP_c+FP_c}$ and $R_c = \frac{TP_c}{TP_c+FN_c}$. The **Binary-F1** score is the harmonic mean of precision and recall for the positive class:

$$\text{F1}_c = 2 \cdot \frac{P_c \cdot R_c}{P_c + R_c} \tag{2}$$

To address class imbalance, we employ **Macro-F1** (unweighted mean of F1 across all $|\mathcal{C}|$ classes) and **Micro-F1** (F1 computed over globally aggregated TP, FP, and FN):

$$\text{Macro-F1} = \frac{1}{|\mathcal{C}|}\sum_{c=1}^{|\mathcal{C}|} \text{F1}_c, \quad \text{Micro-F1} = \frac{2\sum_c TP_c}{2\sum_c TP_c + \sum_c FP_c + \sum_c FN_c} \tag{3}$$

ROC-AUC. The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) evaluates the model's discriminative ability across all decision thresholds. Mathematically, it is the probability that a randomly chosen positive sample x^+ is assigned a higher predicted probability $\hat{p}$ than a randomly chosen negative sample x^-:

$$\text{ROC-AUC} = \mathbb{E}_{x^+,x^-}\left[\mathbb{I}(\hat{p}(x^+) > \hat{p}(x^-))\right] \tag{4}$$

B.1.2 Set-based Metrics (Multiple Valid Outputs)

For tasks yielding a set of elements (e.g., adverse drug event sets), let Y denote the ground truth set and $\hat{Y}$ denote the predicted set.

Exact Match (EM) and Partial Match. **Exact Match** requires the predicted set to be strictly identical to the target set:

$$\text{EM} = \mathbb{I}(\hat{Y} = Y) \tag{5}$$

Partial Match is quantified using the Jaccard Similarity, which measures the degree of overlap:

$$\text{Partial Match (Jaccard)} = \frac{|\hat{Y} \cap Y|}{|\hat{Y} \cup Y|} \tag{6}$$

Set-F1. Set-F1 adapts the standard F1-score for unranked sets. We define Set-Precision $P_{set} = \frac{|\hat{Y} \cap Y|}{|\hat{Y}|}$ and Set-Recall $R_{set} = \frac{|\hat{Y} \cap Y|}{|Y|}$. The Set-F1 is formulated as:

$$\text{Set-F1} = 2 \cdot \frac{P_{set} \cdot R_{set}}{P_{set} + R_{set}} = \frac{2|\hat{Y} \cap Y|}{|\hat{Y}| + |Y|} \tag{7}$$

B.1.3 Generation Metrics (Open-ended Reasoning)

For text generation tasks, let $\boldsymbol{r} = (r_1, ..., r_L)$ be the reference sequence and $\boldsymbol{c} = (c_1, ..., c_M)$ be the generated candidate sequence.

BLEU and ROUGE. **BLEU** evaluates precision-based n-gram overlap penalized by a brevity penalty (BP):

$$\text{BLEU} = \text{BP} \cdot \exp\left(\sum_{n=1}^{N} w_n \log p_n\right), \quad \text{where BP} = \begin{cases} 1 & \text{if } M > L \\ e^{1-L/M} & \text{if } M \leq L \end{cases} \tag{8}$$

where p_n is the modified n-gram precision. Conversely, **ROUGE** (specifically ROUGE-L) focuses on recall via the Longest Common Subsequence (LCS):

$$R_{lcs} = \frac{|\text{LCS}(\boldsymbol{r}, \boldsymbol{c})|}{L}, \quad P_{lcs} = \frac{|\text{LCS}(\boldsymbol{r}, \boldsymbol{c})|}{M}, \quad \text{ROUGE-L} = \frac{(1+\beta^2) R_{lcs} P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}} \tag{9}$$

LLM-as-a-judge. To capture semantic consistency beyond lexical overlap, we define an evaluation function $\mathcal{J}_\theta$ parameterized by a strong language model. Given an input query q, the reference answer $\boldsymbol{r}$, and the candidate answer $\boldsymbol{c}$, the judge yields a binary or scalar score s:

$$s = \mathcal{J}_\theta(\text{prompt\_template}(q, \boldsymbol{r}, \boldsymbol{c})) \tag{10}$$

where $s \in [0, 1]$ represents semantic alignment, factual correctness, and reasoning coherence.

To improve evaluation consistency, the judge operates under structured prompts with task-specific checklists and constrained outputs. Detailed calibration strategies and reliability considerations are provided in Appendix C.

B.2 Analysis of Polypharmacy Prediction Trends

As shown in Figure 7, most models exhibit stable or improving performance as the number of drugs increases, suggesting that LLMs can capture aggregated interaction patterns at the outcome level. However, this trend does not necessarily reflect true compositional reasoning, as higher scores may arise from memorized co-occurrence patterns rather than mechanistic understanding. Notably, several models show performance saturation or fluctuation under higher drug counts, indicating limited robustness in handling complex multi-drug interactions. These results further highlight the gap between predictive performance and reliable reasoning in polypharmacy settings.

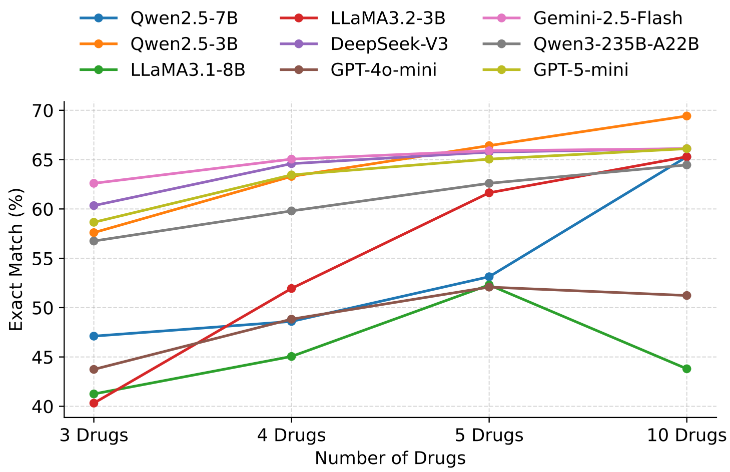

Figure 7: Multiple-drug prediction performance under increasing polypharmacy complexity. Exact Match scores indicate that models can capture outcome-level interactions, but their performance does not necessarily reflect mechanistic or evidence-based reasoning.

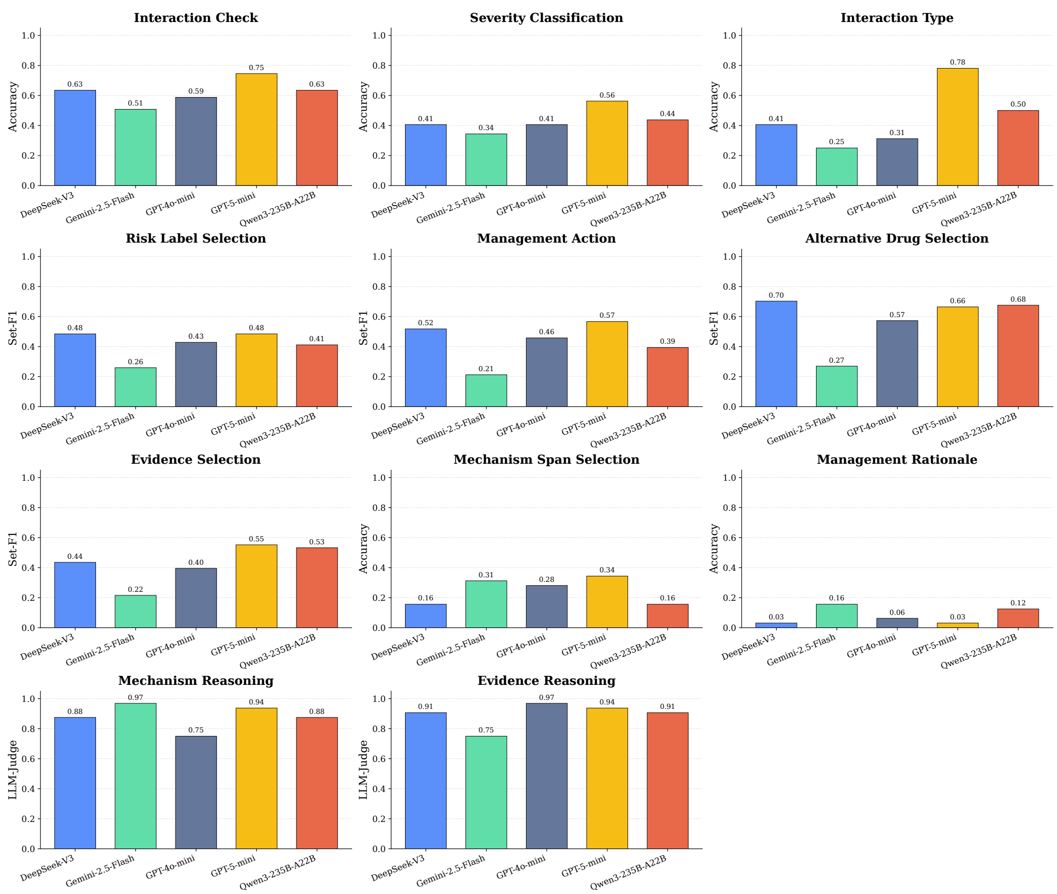

Figure 8: Fine-grained evaluation of LLMs on SafeDrug-small across objective, semi-structured, and generative tasks. While models perform well on interaction checking and generation-based reasoning, performance drops significantly on evidence selection and mechanism span alignment, highlighting a gap between fluent generation and evidence-grounded reasoning.

B.3 Fine-grained Task Analysis on SafeDrug-small

We present a fine-grained evaluation across all task categories in SafeDrug-small, including objective, semi-structured, and generative tasks. Figure 8 reports model performance on representative tasks such as interaction check, severity classification, interaction type prediction, risk label selection, management action prediction, alternative drug selection, evidence selection, mechanism span selection, and reasoning-based tasks.

Objective Tasks. For objective tasks such as interaction check and alternative drug selection, most models achieve relatively higher performance, indicating that LLMs can capture explicit interaction patterns and structured decision rules. However, performance on severity classification and interaction type prediction remains moderate, suggesting limitations in distinguishing finer-grained clinical attributes.

Semi-structured Tasks. Performance drops noticeably on semi-structured tasks, including evidence selection, mechanism span selection, and management rationale selection. In particular, mechanism span selection and rationale prediction show substantial degradation, reflecting the difficulty of aligning outputs with precise spans or structured evidence. This suggests that LLMs struggle to ground their predictions in specific supporting information, even when relevant context is provided.

Generative Reasoning Tasks. In contrast, models achieve high scores on mechanism reasoning and evidence reasoning when evaluated with LLM-as-a-judge. This indicates that models can produce fluent and seemingly plausible explanations. However, combined with the weaker performance on evidence selection and span alignment, this suggests that such explanations may not be faithfully grounded in the provided evidence, highlighting a gap between generation quality and factual grounding.

Key Observations. Overall, the results reveal a consistent pattern: LLMs perform well on surface-level prediction and free-form generation, but struggle with tasks requiring precise evidence alignment and structured reasoning. This discrepancy indicates that current models may rely on memorized patterns or heuristic reasoning rather than truly grounded, mechanism-aware inference.

C Evaluation Protocol Clarifications

To improve transparency and reproducibility, we provide additional details on key evaluation design choices, including error annotation, LLM-based judgment, and metric definitions.

C.1 Error Taxonomy and Annotation Strategy

Sampling Strategy. To conduct a robust qualitative error analysis, we designed a stratified random sampling strategy. We first aggregated all incorrect predictions (where Exact Match = 0) from the evaluated models across all datasets. These errors were then mapped into five macro-task families: *Prediction*, *Risk Assessment*, *Evidence Grounding*, *Mechanistic Reasoning*, and *Compositional Reasoning*. Using a fixed random seed to ensure reproducibility, we sampled up to **80 error instances per task family**, yielding a total error analysis pool of approximately 400 samples.

Annotation Process. To scale error attribution across this diverse pool, we adopted an **LLM-as-an-annotator** approach. The annotator is prompted under a strict expert clinical pharmacologist persona and provided with the question, ground truth, model prediction, and definitions of five mutually exclusive error categories (e.g., *Hallucination, Missed Signal, Over-Risk, Context Misalignment, Superficial Reasoning*).

Reliability Considerations. We ensure consistency through: (1) deterministic prompting with fixed templates, (2) mutually exclusive category design, (3) context-grounded justification requirements, and (4) fixed random seeds for reproducibility. This enables stable and scalable error attribution for macro-level analysis.

C.2 Calibration of LLM-as-a-Judge

For generative tasks, we employ an LLM-based judge to evaluate free-text outputs.

Calibration. The judge is constrained using **task-specific checklists** (e.g., causal explanation, evidence usage, clinical relevance) and grounded on both the reference answer and the source context. It outputs a strict binary score (1: correct/aligned, 0: incorrect or hallucinated), converting evaluation into deterministic checklist verification.

Reliability Considerations. We improve robustness via: (1) checklist-constrained evaluation, (2) explicit context grounding, (3) binary decision space, and (4) standardized prompt templates. These constraints reduce arbitrariness and improve reproducibility.

D Data Licenses and Redistribution Permissions

To answer the concerns regarding data redistribution, we confirm that all original data sources incorporated in our study—including CT-ADE[3], DDInter2.0[4], HODDI[5], SIDER[6], OFFSIDES[7], DrugBank[8], and the pediatric resources[9]—are publicly accessible. Their respective licensing terms explicitly permit usage for academic research and non-commercial purposes.

In strict adherence to the redistribution requirements of these foundational resources, we guarantee that all derived instances, instruction-tuning datasets, and generated evidence constructed during this study will be fully open-sourced. They will be officially released to the community under a clear and standard non-commercial license (e.g., Creative Commons Attribution-NonCommercial 4.0 International, CC BY-NC 4.0). This ensures that our derived resources remain freely available to support future academic and non-commercial research endeavors in the medical domain.

E Related Work

Drug–Drug Interaction and Drug Safety Modeling. Drug–drug interaction (DDI) prediction has been extensively studied using machine learning and graph-based methods, aiming to identify potential adverse interactions or beneficial drug combinations. Existing approaches typically model DDI data as graphs and leverage drug features [36, 37, 38], knowledge graph embeddings [39, 40, 41], or graph neural networks [11, 42, 43] to predict interaction types. While these methods have achieved strong predictive performance, their evaluation is largely limited to classification settings, focusing on accuracy rather than a deeper understanding of pharmacological mechanisms or safety implications.

Benchmarking and Evaluation for DDI Prediction. Recent work has recognized the limitations of current evaluation protocols and proposed more systematic benchmarking efforts [7, 15]. For example, prior studies highlight the lack of unified evaluation frameworks, inconsistencies in datasets and metrics, and insufficient consideration of real-world scenarios such as interactions involving new drugs [13, 14]. Furthermore, performance can vary significantly across different DDI types and data distributions, and the role of side information (e.g., biomedical networks) remains insufficiently understood. These findings suggest that existing benchmarks primarily capture predictive capability but fall short in assessing robustness, generalization, and practical applicability.

Limitations of Existing Benchmarks for Pharmacological Reasoning. Despite progress in both DDI prediction and LLM evaluation, there remains a fundamental gap between current benchmarks and the requirements of real-world drug safety assessment. Existing DDI benchmarks emphasize prediction accuracy but do not evaluate reasoning, evidence grounding, or clinical decision support. Conversely, LLM benchmarks focus on general QA and lack domain-specific structure for pharmacological reasoning. Critically, current evaluation frameworks do not account for multi-drug

[3]https://huggingface.co/anthonyyazdaniml

[4]https://ddinter2.scbdd.com/

[5]https://github.com/TIML-Group/HODDI

[6]http://sideeffects.embl.de

[7]https://nsides.io/#offsides-and-twosides

[8]https://go.drugbank.com/

[9]https://github.com/TY-CSU/Predicting-Adverse-Drug-Reactions-in-Children

interactions, population heterogeneity, or the hierarchical nature of drug safety reasoning. As a result, the ability of LLMs to support safety-critical pharmacological reasoning remains largely underexplored.

F Broader Impact

SafeDrug aims to advance research on reliable and interpretable pharmacological reasoning in large language models. By providing a large-scale benchmark with evidence grounding, mechanistic annotations, and population-aware evaluation, our dataset can support the development of safer AI systems for biomedical and clinical applications. We hope SafeDrug will facilitate research on robust reasoning, trustworthy evidence utilization, and clinically informed decision support, particularly in safety-critical settings such as adverse drug event prediction and drug–drug interaction analysis.

At the same time, our work also carries potential risks. Models evaluated or trained on SafeDrug may generate plausible but incorrect pharmacological recommendations, especially in complex clinical scenarios involving polypharmacy, rare populations, or incomplete evidence. Such outputs could be harmful if used directly in real-world clinical decision-making without expert supervision. In addition, biases inherited from existing biomedical databases and literature may lead to uneven performance across demographic groups or disease conditions.

To mitigate these risks, SafeDrug is released strictly for research and benchmarking purposes rather than clinical deployment. We explicitly emphasize that benchmark outputs should not be interpreted as medical advice or treatment recommendations. Furthermore, we include evidence-grounded tasks and population-aware evaluation to encourage future research on transparency, fairness, and reliability in biomedical LLM systems.

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